surface tension ↑ → ปอดขยายตัวได้ยากขึ้น

D. Increased chest wall elastic recoil

E. Increased transpulmonary pressure

19. Which of following pathologies reduced lung compliance?

A. Asthma

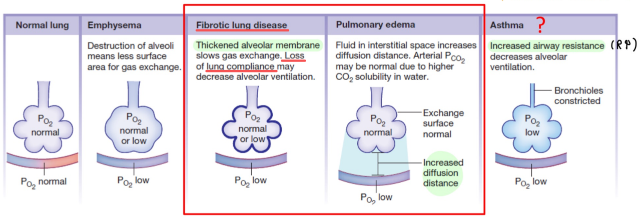
B. Emphysema

C. Chronic bronchitis

[✓ D.] Pulmonary fibrosis

E. Chronic obstructive pulmonary disease

? มีผลตรงไหนต่อ Fick'


Normal lung
Emphysema
Fibrotic lung disease
Pulmonary edema
Asthma
?
(R↑)
P_{O_2} normal
P_{O_2} normal
P_{O_2} normal or low
P_{O_2} low
P_{O_2} normal or low
P_{O_2} low
P_{O_2} normal
Exchange surface normal
Increased diffusion distance
P_{O_2} low
Bronchioles constricted
P_{O_2} low
P_{O_2} low

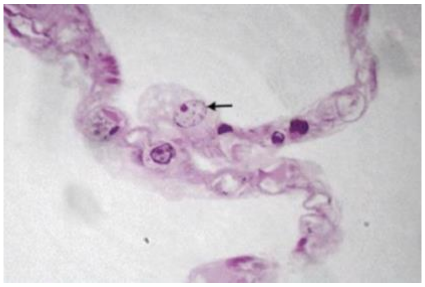
Emphysema: Destruction of alveoli means less surface area for gas exchange.

Fibrotic lung disease: Thickened alveolar membrane slows gas exchange. Loss of lung compliance may decrease alveolar ventilation.

Pulmonary edema: Fluid in interstitial space increases diffusion distance. Arterial P_{CO_2} may be normal due to higher CO_2 solubility in water.

Asthma: Increased airway resistance decreases alveolar ventilation.

20. In a hematoxylin & eosin-stained section of the interalveolar septum, as shown in the picture. Which of the following is the function of the pointed cell?



น่าจะเป็น dust cell

A. Angiotensin converting enzyme production

B. Mucus secretion

C. Gas exchange – type I pneumocyte

[✓ D.] Phagocytosis

E. Surfactant production –type II pneumocyte

21.ส่วนไหนมี airway resistance สูงสุดขณะหายใจปกติ

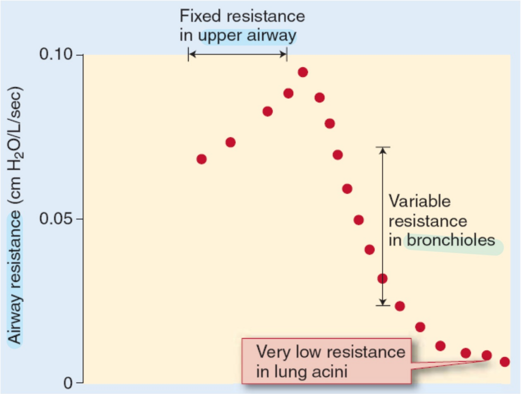
[✓ A.] Medium size bronchiole

B. Alveolar duct

C. –

D. –

E. –



22. Which of the following causes a decrease of airway resistance?

A. Allergic attack ↑R

B. Asthmatic attack ↑R

[✓ C.] Albuterol inhalation β_2-adrenergic agonist → bronchodilation → ↑ϕ → ↓R

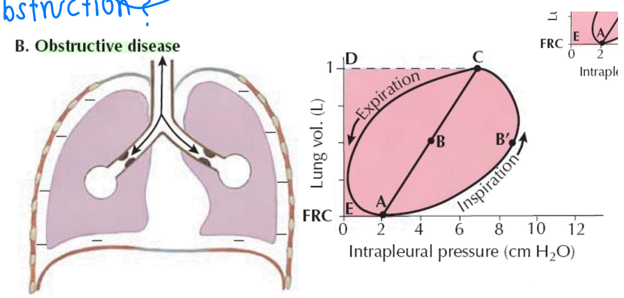
D. Vagal activation ↑R

E. Bronchoconstriction ↑R

ไม่แน่ใจ TwT

23. Which of the following increase resistant work of breathing

lower airway obstruction?



B. Obstructive disease
FRC
E
A
0
2
Intraple
1
D
C
Expiration
B
B′
Inspiration
Lung vol. (L)
FRC
E
A
0
2
4
6
8
10
12
Intrapleural pressure (cm H₂O)

[✓ A.] Pneumonia

B. Tuberculosis

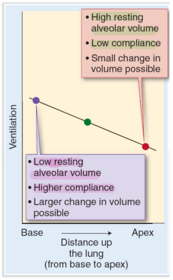
C. Emphysema X

D. Fibrosis X

E. –

24. In the upright healthy lung. Which of the following has a higher value at the apex than base

Ventilation ที่ base > apex



A. Alveolar pressure เท่ากัน ?

B. Alveolar ventilation ↓

C. Alveolar compliance ↓

D. Intrapleural pressure ↓

[✓ E.] Alveolar transpulmonary pressure

25. What is the cause that expand perfusion Zone 1?

$P_A > P_a > P_v$

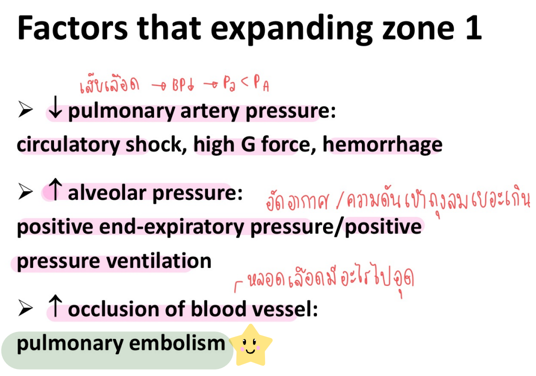
A. Blood infusion

B. Supine position

[✓ C.] Pulmonary embolism

D. Increased arterial blood pressure

E. Decreased transpulmonary pressure



Factors that expanding zone 1

เสียเลือด → BP↓ → $P_a < P_A$

- ↓ **pulmonary artery pressure:**

circulatory shock, high G force, hemorrhage

- ↑ **alveolar pressure:** อัดอากาศ / ความดันเข้าถุงลมเยอะเกิน

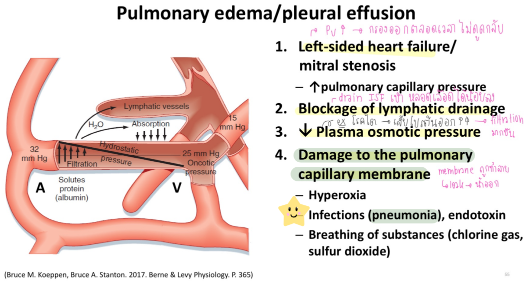
positive end-expiratory pressure/positive pressure ventilation

หลอดเลือดมีอะไรไปอุด

- ↑ **occlusion of blood vessel:**

pulmonary embolism

26. Which of the following is the underlying mechanism of the pulmonary edema in case of pneumonia?


32 mm Hg
Lymphatic vessels
H_2O
Absorption
15 mm Hg
Hydrostatic pressure
Filtration
25 mm Hg
Oncotic pressure
Solutes protein (albumin)
A
V

Pulmonary edema/pleural effusion

↱ Pv ↑ → กรองออกตลอดเวลา ไม่ดูดกลับ

1. **Left-sided heart failure/ mitral stenosis**
   - **↑pulmonary capillary pressure**

↱ drain ISF เข้า หลอดเลือดได้น้อยลง

2. **Blockage of lymphatic drainage**

↱ ex โรคไต → เสียโปรตีนออก ↑↑ → filtration มากขึ้น

3. **↓ Plasma osmotic pressure**
4. **Damage to the pulmonary capillary membrane** membrane ถูกทำลาย ↳ leak → น้ำออก
   - **Hyperoxia**
   - **Infections (pneumonia), endotoxin**
   - **Breathing of substances (chlorine gas, sulfur dioxide)**

(Bruce M. Koeppen, Bruce A. Stanton. 2017. Berne & Levy Physiology. P. 365)

55

A. Increased plasma oncotic pressure

[✓ B.] Increased lung ~~capacity~~ capillary permeability

C. Increased plasma hydrostatic pressure

D. Decreased plasma oncotic pressure

E. Decreased lymphatic drainage of the visceral pleura

27. Which of the following conditions causes an increase in ventilation/perfusion (V/Q) ratio?

A. Atelectasis ↓V

B. Asthmatic attack ↓V

C. Airway obstruction ↓V

D. Pulmonary edema ↓V

[✓ E.] Pulmonary embolism ↓P

28.-

29.-

30. Which of the following causes a decrease in arterial PO2 in asthmatics patients?

↳ ↑R → ↓ΔP → diffusion ↓

A. Increased alveolar PCO2 X

B. Increased pulmonary blood flow X

C. Decreased alveolar surface area X

D. Increased respiratory membrane thickness X

[✓ E.] Decreased alveolar-capillary PO2 difference

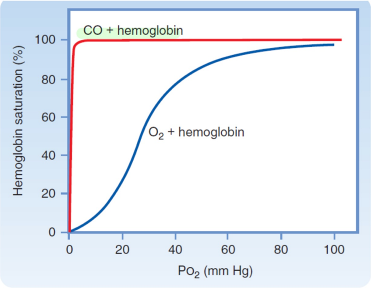
31.-

32.-

33.-

34. Which of the following is alter affinity of hemoglobin for oxygen?

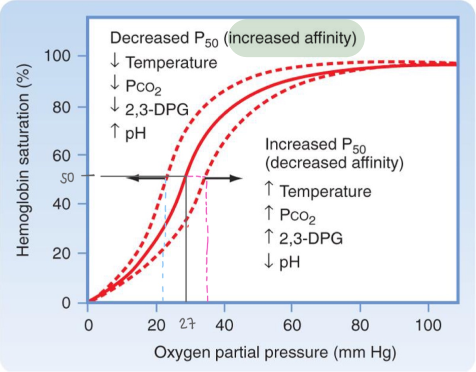
A. Hemoglobin A
B. Ferrous cation
C. Nitrous oxide
[✓ D.] Carbon monoxide
E. Methemoglobin reductase



- **CO poisoning**
    - CO จับ Hb ตำแหน่งเดียวกับ O2 แต่จับได้ดีกว่า 200 เท่า→ ไม่เหลือ Hb จับ O2→ ขาด O2

35. In the oxygen-hemoglobin dissociation curve, which of the following factors cause an increase in oxygen-hemoglobin affinity?

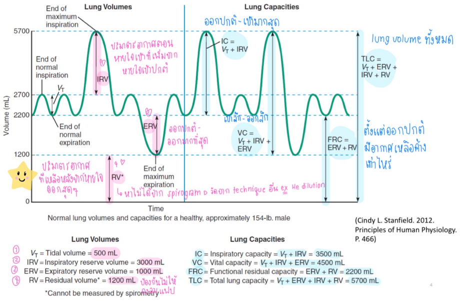
[✓ A.] Sleeping
B. Acidosis X
C. During exercise X
D. Chronic bronchitis X
E. Staying at high altitude X



36.-

37. What is volume of gas when maximum inspiration?

[✓ A.] TLC
B. RV
C. IRV
D. ERV
E. VC



Normal lung volumes and capacities for a healthy, approximately 154-lb. male

(Cindy L. Stanfield. 2012. Principles of Human Physiology. P. 466)

Lung Volumes

① V_T = Tidal volume = 500 mL
② IRV = Inspiratory reserve volume = 3000 mL
③ ERV = Expiratory reserve volume = 1000 mL
④ RV = Residual volume* = 1200 mL

*Cannot be measured by spirometry

Lung Capacities

IC = Inspiratory capacity = V_T + IRV = 3500 mL
VC = Vital capacity = V_T + IRV + ERV = 4500 mL
FRC = Functional residual capacity = ERV + RV = 2200 mL
TLC = Total lung capacity = V_T + ERV + IRV + RV = 5700 mL

38.สิ่งที่คนเป็น emphysema มีมากกว่าคนปกติคืออะไร

↑ FRC, ↑ RV , elastic recoil ↓

[✓ A.] Residual volume
B. –
C. FEV1
D. –
E. –

39.-

40.ให้ rr , dead space กับ total lung มา ให้หาalveolar pressure $V_A = (V_T - V_D) \times f$

41.-

42. Which brain respiratory area generate signal that result in forced expiration during exercise?

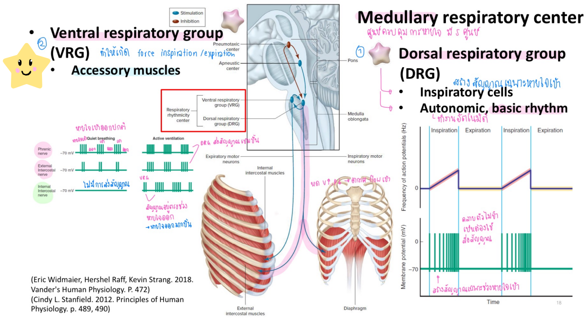
A. APN

B. PNC

C. DRG

[✓ D.] VRG

E. RAS



- **Ventral respiratory group (VRG)** ② ทำให้เกิด force inspiration /expiration
  - **Accessory muscles**

Medullary respiratory center ศูนย์ควบคุมการหายใจ มี 5 ศูนย์

① **Dorsal respiratory group (DRG)** สร้างสัญญาณเฉพาะหายใจเข้า

- **Inspiratory cells**
- **Autonomic, basic rhythm** ทำงานอัตโนมัติ

Stimulation
Inhibition
Pneumotaxic center
Apneustic center
Pons
Respiratory rhythmicity center
Ventral respiratory group (VRG)
Dorsal respiratory group (DRG)
Medulla oblongata
Expiratory motor neurons
Inspiratory motor neurons
Internal intercostal muscles
External intercostal muscles
Diaphragm
หายใจเข้าออกปกติ
Quiet breathing
Active ventilation
DRG ส่งสัญญาณเพิ่มขึ้น
Phrenic nerve
External intercostal nerve
Internal intercostal nerve
–70 mV
ไม่มีการส่งสัญญาณ
VRG
สัญญาณอยู่ตรงช่วงหายใจออก → หายใจออกมากขึ้น
Inspiration
Expiration
Frequency of action potentials (Hz)
Membrane potential (mV)
0
–70
Time
คลายตัวไม่จำเป็นต้องใช้ส่งสัญญาณ
สร้างสัญญาณเฉพาะช่วงหายใจเข้า
18

(Eric Widmaier, Hershel Raff, Kevin Strang. 2018. Vander's Human Physiology. P. 472)
(Cindy L. Stanfield. 2012. Principles of Human Physiology. p. 489, 490)

43. Deep and slow breathing patterns would occur in a person who has brainstem transection in?

A. Above midbrain

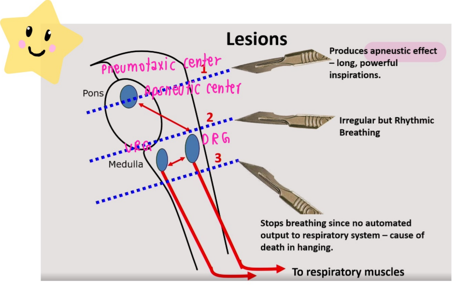
B. Upper pons

C. Below medulla

[✓ D.] Between pons and medulla

E. Between midbrain and pons

▼ Key



Lesions
Pneumotaxic center
Apneustic center
Pons
Medulla
VRG
DRG
1
2
3
Produces apneustic effect – long, powerful inspirations.
Irregular but Rhythmic Breathing
Stops breathing since no automated output to respiratory system – cause of death in hanging.
To respiratory muscles

44. Which effects will occur when above pons has been complete transducted?

[✓ A.] The voluntary control of breathing disappears

B. The Hering-Breuler reflex will abolish

C. Only inhalation occurs

D. Peripheral chemoreceptor will not response

E. Central chemoreceptor will not response

45. What of fallowing receptors is responsible for dyspnea in a patient with left sided heart failure?

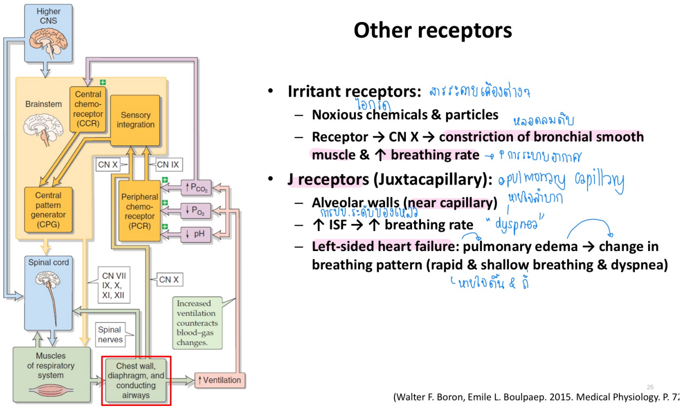
[✓ A.] J receptor

B. Proprioceptor

C. Irritant receptor

D. Central chemoreceptor

E. Peripheral chemoreceptor


Higher CNS
Brainstem
Central chemo-receptor (CCR)
Sensory integration
Central pattern generator (CPG)
CN X
CN IX
Peripheral chemo-receptor (PCR)
↑P_{CO_2}
↓P_{O_2}
↓ pH
Spinal cord
CN VII IX, X, XI, XII
CN X
Spinal nerves
Increased ventilation counteracts blood–gas changes.
Muscles of respiratory system
Chest wall, diaphragm, and conducting airways
↑Ventilation

Other receptors

- Irritant receptors: สารระคายเคืองต่างๆ
  - Noxious chemicals & particles ไอกรด
  - Receptor → CN X → constriction of bronchial smooth muscle & ↑ breathing rate หลอดลมตีบ → ↑ การระบายอากาศ
- J receptors (Juxtacapillary): อยู่ pulmonary capillary
  - Alveolar walls (near capillary) หายใจลำบาก
  - ↑ ISF → ↑ breathing rate การปป.ระดับของเหลว "dyspnea"
  - Left-sided heart failure: pulmonary edema → change in breathing pattern (rapid & shallow breathing & dyspnea) หายใจตื้น & ถี่

25

(Walter F. Boron, Emile L. Boulpaep. 2015. Medical Physiology. P. 7

46. Which receptor is responsible for increased breathing rate ในคนเป็น DM?

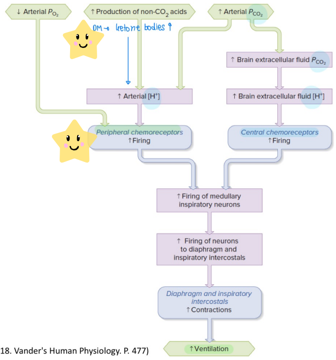
A. J receptor

B. Proprioceptor

C. Irritant receptor

D. Central chemoreceptor

[✓ E.] Peripheral chemoreceptor


↓ Arterial P_{O_2}
↑ Production of non-CO_2 acids
↑ Arterial P_{CO_2}
DM → ketone bodies ↑
↑ Brain extracellular fluid P_{CO_2}
↑ Arterial [H^+]
↑ Brain extracellular fluid [H^+]
Peripheral chemoreceptors ↑Firing
Central chemoreceptors ↑Firing
↑ Firing of medullary inspiratory neurons
↑ Firing of neurons to diaphragm and inspiratory intercostals
Diaphragm and inspiratory intercostals ↑Contractions
↑Ventilation

in Strang. 2018. Vander's Human Physiology. P. 477)

47. -

48. -

49. 18-years-old man got a fever and screening test was positive for covid-19 infection. He present productive cough with phlegm. Which of the following mucolytic drug used for relieved cough.

A. Codeine - centrally acting antitussive drug

[✓ B.] N-acetylcysteine

C. Benzonatate - peripheral acting antitussive drug

D. Chlorpheniramine - antihistamine

E. Dextometrophan - centrally acting antitussive drug

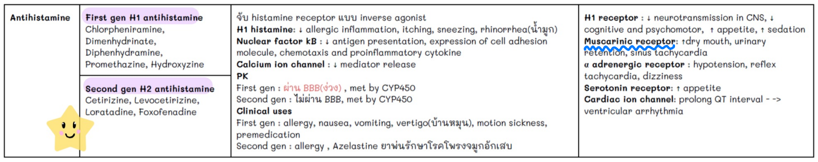
50. patient 45 yrs using Chlorpheniramine have side effect of dry mouth and difficulty urination, explain mechanism of chlorpheniramine in causing dry mouth and difficulty urination?

A. .... nicotinic receptor

[✓ B.] ... muscarinic receptors

C. ... beta adrenergic receptors

D. .... alpha 1 adrenergic receptors



Antihistamine	**First gen H1 antihistamine** Chlorpheniramine, Dimenhydrinate, Diphenhydramine, Promethazine, Hydroxyzine	จับ histamine receptor แบบ inverse agonist **H1 histamine**: ↓ allergic inflammation, itching, sneezing, rhinorrhea(น้ำมูก) **Nuclear factor kB** : ↓ antigen presentation, expression of cell adhesion molecule, chemotaxis and proinflammatory cytokine **Calcium ion channel** : ↓ mediator release **PK** First gen : ผ่าน BBB(ง่วง) , met by CYP450 Second gen : ไม่ผ่าน BBB, met by CYP450 **Clinical uses** First gen : allergy, nausea, vomiting, vertigo(บ้านหมุน), motion sickness, premedication Second gen : allergy , Azelastine ยาพ่นรักษาโรคโพรงจมูกอักเสบ	**H1 receptor** : ↓ neurotransmission in CNS, ↓ cognitive and psychomotor, ↑ appetite, ↑ sedation **Muscarinic receptor**: ↑dry mouth, urinary retention, sinus tachycardia **α adrenergic receptor** : hypotension, reflex tachycardia, dizziness **Serotonin receptor**: ↑ appetite **Cardiac ion channel**: prolong QT interval - -> ventricular arrhythmia
	Second gen H2 antihistamine Cetirizine, Levocetirizine, Loratadine, Foxofenadine		

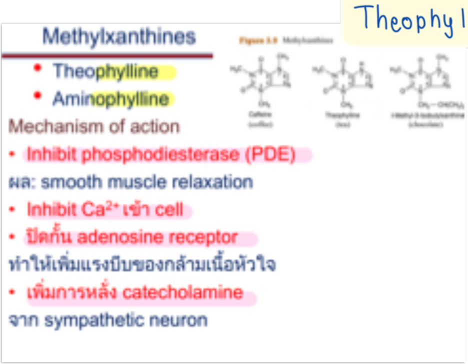
E. irgonist of H1 receptor

51. -

52. neonate birth at 34 gestation weeks presented with apnea. Which drug mechanism stimulates central nervous system

Theophyline

A. agonist beta 2
B. agonist adenosine receptor
[✓ C.] antagonist adenosine receptor
D. antagonist muscarinic receptor
E. agonist g protein coupled receptor


Methylxanthines
- Theophylline
- Aminophylline

Mechanism of action
- Inhibit phosphodiesterase (PDE)

ผล: smooth muscle relaxation
- Inhibit Ca^{2+} เข้า cell

√
- ปิดกั้น adenosine receptor

ทำให้เพิ่มแรงบีบของกล้ามเนื้อหัวใจ
- เพิ่มการหลั่ง catecholamine

จาก sympathetic neuron

53. A 65 year-old male with chronic obstructive pulmonary disease (COPD). He has been smoking for 40 years. He develops dyspnea, orthopnea and cough with phlegm. Which of following is used to relieve dyspnea and decrease secretion?

A. Oral prednisolone
B. Oral theophylline
C. Inhaled budesonide
D. Inhaled salbutamol
[✓ E.] Inhaled ipratropium bromide = muscarinic antagonist

54. -

55. Which is following conditions is a contraindications for oral albuterol

selective β2 agonist

A. Anemia
B. Cataract
C. Peptic ulcer
[✓ D.] Hypokalemia
E. Chronic Cough

56. which drug has FC epsilon 3 Receptor as target

= IgE

[✓ A.] Omalizumab
B. Zileuton
C. Theophylline
D. Montelukast
E. -

57. A 67-year-old female with COPD suffered from dyspnea on exertion. Her symptom is not improved after given with ipratropium and salbutamol. Which of the following drug is appropriate alternative bronchodilator?

[✓ A.] Theophylline ใช้ยา ตอนที่ให้ B_2 agonist หรือ salbutamol ไม่ได้ผล
B. Metoprolol
C. Zileuton
D. Beclomethasone
E. Salmeterol

58. -

59. -

60. Which condition should be "heterogeneous opacity" in Chest x-ray

= มีของเหลวสะสมใน Alveoli , Patchy infiltration

A. Hemothorax
[✓ B.] Pulmonary hemorrhage
C. Pleural effusion
D. Thymoma
E. Atelectasis

61. -

62. -

63. Which of the following is the common abnormality that associated with pulmonary hypoplasia?

A. Pulmonary agenesis
B. HMD
C. CHD
[✓ D.] diaphragmatic hernia
E. Intrauterine infection

64. What are antibodies of immunohistochemistry strain, using confirm small cell lung carcinoma in the lung?

A. CK7 and CK20
B. CK7 and TTF-1
C. TTF-1 and CK20
[✓ D.] Chromogranin and TTF-1
E. Chromogranin and synaptophysin

65. Bronchus มี thick fibrous membrane , Dilated bronchial lumen from proximal to distal part , Yellow green secretion in lumen

A. Atelectasis
B. Bronchomalacia
C. Bronchial Asthma
D. acute bronchitis
[✓ E.] chronic bronchitis

66. -

67. พบได้ใน asthma แต่ไม่เจอใน chronic bronchitis

[✓ A.] Infiltration eosinophil
B. Thick basements membrane
C. Thick mucus
D. -
E. -